FLAGSHIP 04 · PERSONAL THERMAL MANAGEMENT SERIES

LC Frost-PCM/A

PCM Cooling Vest (Transparent TPU Edition) · Bio-Based C17 Phase Change

Bio-Based Phase Change Armor. Constant 22.08°C Torso Comfort.

Built with high-energy-density Phase Change Material (PCM) that absorbs body heat at a constant comfort temperature (~23°C), the LC Frost-PCM/A delivers gentle, long-lasting torso cooling. A single cold-storage cycle provides 1-2 hours of relief in 35°C high-heat environments. Reusable thousands of times - an intrinsically safe, wire-free personal micro-climate.



01 Gentle Human-Comfort Cooling

Operates at 22.08°C, completely preventing skin vasoconstriction or frostbite risks caused by traditional 0°C ice packs.

02 100% Passive & Intrinsically Safe

Requires no batteries, fans, or wires. Zero electrical hazards in cleanrooms, hazardous gas zones, and wet areas.

03 Zero-Degradation C17 Formulation

Pure bio-based C17 n-alkane PCM withstands thousands of freeze-thaw cycles without performance loss.

04 Plant-Derived & Biodegradable

100% biomass formulation, non-petroleum derived, environmentally safe and non-toxic.

KEY METRIC 01
22.08°C
Constant Phase Change Comfort Temperature

KEY METRIC 02
225 J/g
High Latent Heat of Fusion

KEY METRIC 03
0.55kg
Ultra-Lightweight Transparent TPU Chassis

KEY METRIC 04
1-2 Hours
Continuous Cooling per Charge Cycle

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



100% BIO-BASED C17 PCM · 225 J/G

Bio-Based C17 n-Alkane Phase Change Matrix

Plant-derived C17 phase change crystals transition between solid and liquid at an exact 22.08°C plateau, absorbing 225 J/g of metabolic heat without causing cold shock or condensation dampness.



HIGH-STRENGTH TRANSPARENT TPU

High-Strength Medical-Grade TPU RF Welded Seams

High-frequency welded Thermoplastic Polyurethane (TPU) grid channels resist puncture, chemical splashes, and biological contamination for sterile medical and cleanroom deployment.



DUAL SIDE-RELEASE TACTICAL BUCKLES

Universal Dual Side-Release Webbing System

Low-profile acetal buckles and heavy-duty webbing secure the 0.55kg vest beneath sealed hazmat or cleanroom coveralls with zero bulk.

ANATOMICAL & MECHANICAL SUBSYSTEMS (4 NODES)

1. **Fast-Cooling Grid Matrix [Anterior & Dorsal C17 Cell Matrix]**
   Independent encapsulated PCM cells conform across the chest and back for maximum heat exchange area.

2. **Universal Adjustable Buckles [Lateral Tension Fasteners]**
   Dual side-release black webbing buckles adapt to all body sizes and uniform styles.

3. **Medical-Grade TPU Shell [High-Frequency Welded TPU Encapsulation]**
   Waterproof and oil-resistant transparent film enables instant sanitization and wipe-down cleaning.

4. **Low-Profile Under-Layer Cut [0.55kg Sub-Suit Ergonomic Cut]**
   Ultra-slim 0.55kg profile fits seamlessly under hazmat suits, cleanroom suits, or high-vis vests.

FIELD DEPLOYMENT | SEMICONDUCTOR CLEANROOM & BIO-SAFETY LEVEL PPE DEPLOYMENT



Sealed Cleanrooms, Medical PPE & High-Heat Sanitation Operations

Operators wearing impermeable hazmat suits, medical PPE coveralls, or cleanroom bunny suits cannot rely on evaporative sweat cooling or fan-blown air. At just 0.55kg, the crystal-transparent LC Frost-PCM/A slips cleanly beneath sealed outer garments, maintaining a constant 22.08°C micro-climate for 1-2 hours per cycle and recharging in 60 minutes at -18°C.

22.08°C Constant	225 J/g	1 Hour at -18°C
Phase Change Plateau	Latent Heat Capacity	Fast Freezer Recharge

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Frost-PCM/A (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Frost-PCM/A
IDENTITY & COMPLIANCE	**Color**	Crystal Transparent (customizable PCM dye options)
IDENTITY & COMPLIANCE	**Target Users**	Workers in extreme high-temperature environments, cleanrooms, medical PPE, sanitation
PERFORMANCE & MECHANICS	**Heat of Fusion**	225 J/g
PERFORMANCE & MECHANICS	**Phase Change Temperature**	22.08°C
PERFORMANCE & MECHANICS	**Cooling Duration**	1-2 hours (reference based on labor intensity and ambient temperature)
PERFORMANCE & MECHANICS	**Cold Storage Recharge Time**	Frozen at -18°C for 1 hour, OR refrigerated at 0-4°C for 1-2 hours
MATERIALS & ERGONOMICS	**Shell Material**	High-strength Thermoplastic Polyurethane (TPU)
DIMENSIONS & WARRANTY	**Size**	One Size Fits All
DIMENSIONS & WARRANTY	**Net Weight**	0.55kg
DIMENSIONS & WARRANTY	**Package Dimensions**	40 x 30 x 10cm
DIMENSIONS & WARRANTY	**Optional Accessories**	Custom PCM dye colors
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Frost-PCM/A

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

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